KONGU ENGINEERING COLLEGE

TRANSFORM YOURSELF

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AI POWERED SKIN SCREENING AND REFERRAL SYSTEM

ABSTRACT

Access to **specialized dermatological care** is frequently restricted by **geographic barriers**, **long wait times**, and a **shortage of certified skin specialists**, particularly in rural areas. When individuals observe new or changing **skin abnormalities**, the lack of immediate guidance often leads to **anxiety**, **self-diagnosis errors**, and **delayed medical evaluation**. To address this critical healthcare gap, **DermaVision AI** was developed as a web-based, **AI-assisted preliminary skin screening platform** designed to foster **early awareness**, reduce patient uncertainty, and facilitate **timely professional consultation**.

DermaVision AI delivers a comprehensive suite of digital healthcare features designed to empower patients and streamline clinical workflows. The platform allows users to **capture or upload skin lesion images** for automated preliminary screening across **153 supported condition classes**. When an abnormality is identified, the system generates a detailed **condition-specific screening report** outlining the **confidence score**, **clinical symptoms**, **risk factors**, **management guidance**, **warning signs**, and recommendations for when medical attention is necessary. To prevent false alarms, a dedicated safeguard handles **Normal/Healthy Skin** by providing **preventive care advice** instead of a disease report. Furthermore, the platform incorporates **dynamic multilingual report translation** in **English, Tamil, and Hindi**, alongside an **appointment management module** that connects patients directly with **certified dermatologists** for live consultations.

The technical architecture of **DermaVision AI** is powered by a high-performance full-stack ecosystem. The frontend is built using **React 18, TypeScript, and Vite**, while the backend API is powered by **Python 3.11 and FastAPI**. The core AI diagnostic engine utilizes trained **PyTorch deep-learning neural network ensembles**, combining **EfficientNet-B0 and ResNet50** convolutional architectures for feature extraction and pattern classification. Image preprocessing is handled via **OpenCV and Pillow**, while user authentication and real-time appointment synchronization are managed using **Google Firebase Auth and Firestore NoSQL databases**. Public cloud connectivity is established using secure **Cloudflare HTTPS tunnels**. Crucially, **DermaVision AI** is strictly engineered for **preliminary AI-assisted screening** and does **not provide confirmed medical diagnoses**, requiring **certified dermatologists** for final clinical evaluation.

TEAM MEMBERS:

VIGNESHWARAN L - 25CSR342
VISHVA D - 25CSR352
YOGESHWARAN M - 25CSR361
YUVANESH V - 25CSR363